ORCA

Agentic AI Marine Intelligence Platform

For Indian Fishermen, Coastal Researchers & Disaster-Management Agencies

Project Documentation

Generated: September 2026

Stack: Expo (React Native) · FastAPI · MongoDB · Multi-Agent LLM Orchestration

1. Executive Summary

ORCA is an agentic AI-powered marine intelligence platform built for the entire Indian coastline. It gives fishermen, coastal researchers, and disaster-management agencies natural-language answers about ocean and weather conditions, Potential Fishing Zones (PFZ), maps, and safety alerts — in English, Telugu, and Hindi, by voice or text.

The platform is architected around a 7-agent LLM orchestrator layered on top of a DETERMINISTIC, rule-based safety engine: the AI explains conditions in warm, natural language, but it never decides whether it is safe to sail — that verdict always comes from hard-coded threshold rules evaluated against weather data.

- Full-stack: Expo (React Native) frontend + FastAPI backend + MongoDB database
- India-wide coastal coverage across 6 seeded regions (Bay of Bengal & Arabian Sea)
- Multilingual UI and AI responses: English, Telugu, Hindi
- Voice input (speech-to-text) and voice output (text-to-speech) — the 'Marine Copilot'
- Offline-tolerant: cached responses, safe-default answers when APIs are unreachable
- Interactive MapLibre GL maps with PFZ markers, boundaries, heatmaps, and route plotting
- Resilient LLM/voice provider fallback chains (never a dead feature on quota exhaustion)

2. Problem Statement

Build "ORCA" — an Agentic AI-powered marine intelligence platform for Indian fishermen, coastal researchers, and disaster-management agencies. The app provides natural-language answers about ocean/weather conditions, PFZ (Potential Fishing Zones), maps, and safety alerts.

Key requirements:

- India-wide coastal coverage (not a single city/region)
- Multi-agent AI with deterministic safety rules — AI explains, rules decide
- Multilingual UI + AI responses (English, Telugu, Hindi)
- Offline-tolerant architecture
- Voice input and voice output capabilities

3. Technology Stack

3.1 Frontend

Layer	Technology
Framework	Expo (React Native) 54.0.37, React 19.1.0, RN 0.81.5
Routing	expo-router 6.0.24 (file-based, typed routes)
Maps	MapLibre GL JS bundled locally, rendered in react-native-webview
Animations	react-native-reanimated 4.1.1 + worklets
Voice / Audio	expo-audio, expo-file-system
Location	expo-location
Storage	@react-native-async-storage/async-storage
Charts	Custom SVG ChartWidget (react-native-svg)

Layer	Technology
Language	TypeScript 5.9.3

3.2 Backend

Layer	Technology
Framework	FastAPI 0.110.1, Uvicorn ASGI
Database	MongoDB (Motor async) with 2dsphere geospatial indexes
LLM orchestration	emergentintegrations (LlmChat) + litellm
Primary LLM	Gemini 3.5 Flash (user API key)
Fallback LLM	GPT-5.6 Luna via Emergent Universal Key
Voice (primary)	ElevenLabs scribe_v1 STT, eleven_v3 / multilingual_v2 TTS
Voice (fallback)	OpenAI Whisper-1 & TTS-1 via Emergent Universal Key
Geocoding	OpenStreetMap Nominatim (rate-limited + cached)

4. System Architecture

ORCA follows a modular-monolith backend: orchestrator/ (AI reasoning), orchestrator/rules.py (deterministic safety logic), gis/ (geospatial queries), data_adapters/ (regional mock data), voice/ (STT/TTS), and notifications/ (hazard + geofence alerts).

4.1 High-Level Request Flow

1. User sends a message (typed or transcribed from voice).
2. FastAPI /api/chat rate-limits per user and loads conversation history.
3. Planner Agent (LLM) decomposes intent, language/script, and location.
4. Specialist Agents deterministically gather weather, ocean, PFZ, boundary, route, tide, and alert data.
5. Rules Engine evaluates thresholds to produce SAFE / CAUTION / UNSAFE — never the LLM.
6. Synthesizer Agent (LLM) turns the verdict + data into one natural-language reply, honoring the verdict as a hard constraint.
7. Response (text + verdict + widgets + reasoning trace + citations) is persisted and returned.
8. If voice mode, reply text is also sent to TTS and played back.

4.2 The 7-Agent Orchestrator

#	Agent	Role
1	Planner Agent	LLM: intent, timeframe, language/script, location, datasets
2	Data Discovery Agent	Deterministic: maps intent to required datasets
3	Weather Intelligence Agent	Deterministic: weather snapshot -> Rules Engine verdict
4	Ocean Analytics Agent	Deterministic: correlates SST & chlorophyll
5	Geospatial / Risk Agent	Deterministic: $geoNear PFZ, $geoIntersects geofence, route sampling

#	Agent	Role
6	Reporting/Visualization Agent	Deterministic: builds map/chart widget JSON
7	Synthesizer Agent	LLM: natural localized reply, constrained by verdict

4.3 Deterministic Safety Rules

Factor	Caution	Unsafe
Wind speed	≥ 18 kn	≥ 25 kn
Wave height	≥ 1.8 m	≥ 2.5 m
Lightning probability	≥ 30%	≥ 60%
Active cyclone	—	Any active system -> UNSAFE

4.4 Resilience & Fallback Chains

- Chat LLM: Gemini 3.5 Flash -> GPT-5.6 Luna (Emergent Key) on any failure
- Voice STT: ElevenLabs Scribe -> OpenAI Whisper (Emergent Key)
- Voice TTS: ElevenLabs eleven_v3 -> multilingual_v2 -> OpenAI tts-1 (Emergent Key)
- If both LLM providers fail: templated deterministic safe-default answer in-language
- Frontend caches last-known Alerts snapshot; shows stale-data banner offline

5. Supported Coastal Regions (India-Wide)

Any coastal place a user names is geocoded and snapped to the nearest seeded reference region for weather/PFZ/boundary data, while the LLM can still reason qualitatively about any other Indian coastal location.

Region	State	Sea
Kakinada (default)	Andhra Pradesh	Bay of Bengal
Chennai	Tamil Nadu	Bay of Bengal
Kochi	Kerala	Arabian Sea
Mumbai	Maharashtra	Arabian Sea
Paradip	Odisha	Bay of Bengal
Veraval	Gujarat	Arabian Sea

6. Key Features

6.1 Conversational Marine Copilot (Chat Tab)

- Natural-language Q&A about tides, weather, safety, fishing zones, cyclones, lightning
- Deterministic SAFE / CAUTION / UNSAFE verdict badge on safety-relevant answers
- Inline reasoning trace of which agents contributed to the answer
- Inline citations, MapLibre map widget, and trend charts

- Region switcher: GPS auto-detect, free-text search, or manual pick
- Persistent conversation history restored on relaunch

6.2 Marine Safety Center (Alerts Tab)

- Situation Summary card: severity badge, cyclone banner, wind/wave/lightning metrics
- Animated Radar card tied to current severity
- 7-day trend charts: wind, wave, sea-surface temperature
- Active hazard bulletins and personal notifications
- Pull-to-refresh with offline fallback + stale-data banner

6.3 Saved Locations Tab

- Save named coordinates (manual, GPS, or place search)
- Vessel profile toggle
- One-tap 'Ask in Chat' deep link with saved location's context
- One-tap geofence check against restricted/MPA/EEZ boundaries
- Auto-evaluates hazard bulletins for newly saved locations

6.4 Route Planning

- Point-to-point route safety check sampled into 6 segments
- Flags segments crossing restricted/hazardous boundary polygons
- Recommends the safest reachable Potential Fishing Zone
- Route drawn as color-coded polyline with start/end markers

6.5 Voice Agent ("Marine Copilot" voice mode)

- Tap-to-record mic input with full permission-contract handling
- Speech transcribed via ElevenLabs Scribe (fallback: OpenAI Whisper)
- Transcript sent through the same /api/chat pipeline as typed text
- Reply read aloud via ElevenLabs (fallback: OpenAI TTS)
- Auto-stop recording after 30 seconds; recording/transcribing/speaking states

6.6 Multilingual Support

- UI fully localized in English, Telugu, and Hindi
- AI responses match detected input language or app's language preference
- Detects native script AND casual romanized Telugu/Hindi
- Replies in matching tone (romanized in, romanized-style reply out)

7. API Reference

All backend routes are prefixed with /api.

7.1 Health & Core

Method	Endpoint	Description
GET	/api/	Service health + region coverage summary

Method	Endpoint	Description
GET	/api/health	Simple health check with server timestamp

7.2 Chat & Conversation

Method	Endpoint	Description
POST	/api/chat	Main orchestrator entry point
GET	/api/conversations/{session_id}	Retrieve stored conversation history

7.3 Saved Locations

Method	Endpoint	Description
POST	/api/locations	Save a named location (auto hazard eval)
GET	/api/locations	List saved locations for a user
DELETE	/api/locations/{location_id}	Delete a saved location

7.4 Alerts & Situation Intelligence

Method	Endpoint	Description
GET	/api/alerts	Active hazard bulletins for a region
GET	/api/notifications	Personal notifications
GET	/api/situation	Aggregated Alerts-tab snapshot

7.5 Geospatial & Reference Data

Method	Endpoint	Description
POST	/api/geofence/check	Point-in-polygon breach check
GET	/api/data/boundaries	Bundled maritime boundary layer
GET	/api/data/pfz	Potential Fishing Zone advisory
GET	/api/data/ocean	SST/chlorophyll grid + trends
GET	/api/region	Single region config lookup
GET	/api/regions	Full India-wide region roster
GET	/api/regions/detect	Nearest region for a lat/lon
GET	/api/geocode	Forward-geocode place name -> coordinates

7.6 Voice Agent

Method	Endpoint	Description
POST	/api/voice/transcribe	Speech (audio) -> text

Method	Endpoint	Description
POST	/api/voice/speak	Text -> speech (mp3 bytes)

Rate limiting: chat capped at 20 req/60s per user; voice capped at 15 req/60s per client.

8. Frontend Architecture

8.1 Routing (expo-router, file-based)

```
app/
|-- _layout.tsx        Root layout: fonts, splash, providers, Stack
|-- index.tsx          Redirects '/' -> '/(tabs)'
|-- +html.tsx          Web HTML shell
'-- (tabs)/
    |-- _layout.tsx    Bottom tab bar (Chat / Alerts / Saved)
    |-- index.tsx      Chat tab (ChatScreen)
    |-- alerts.tsx     Marine Safety Center
    '-- saved.tsx      Saved Locations tab
```

8.2 Shared Source (src/)

```
src/
|-- api.ts              Typed API client
|-- theme.ts            Design tokens
|-- context/AppContext.tsx   Global state (lang, session, region, toast)
|-- hooks/useVoiceAgent.ts   Recording/playback state machine
|-- utils/storage/      AsyncStorage/localStorage wrapper
'-- components/
    MessageBubble, MapWidget, ChartWidget, RadarCard, SituationCard,
    AlertCard, VerdictBadge, ReasoningTrace, RegionSwitcher,
    LanguagePicker, StaleBanner
```

8.3 Design System

"Maritime Command Console" dark theme — deep navy-black surfaces, command-blue accent, monospace telemetry text, green/amber/red safety semantics, 8pt spacing grid.

9. Data Model (MongoDB Collections)

Collection	Purpose	Key Fields
conversations	Chat session history	session_id, user_id, messages[]
locations	Saved coordinates	id, name, lat, lon, user_id, is_vessel
notifications	Hazard/geofence alerts	id, user_id, type, severity, kind
boundaries	Maritime boundary polygons	key, name, geometry (2dsphere), restricted
pfz	Potential Fishing Zone points	key, name, location (2dsphere), sst_c, chl_mg_m3
geocode_cache	Nominatim result cache	cache_key, query, results[], TTL 30d

10. Permissions

Permission	Platform	Description Shown to User
Location (when in use)	iOS & Android	"Check sea conditions near your location"
Microphone	iOS & Android	"Ask the Marine Copilot questions by voice"

11. Known Limitations & Mocked Data

- Weather/tide/SST/chlorophyll/PFZ/boundary data are DEMO/MOCK per region, clearly labelled in citations
- Gemini free-tier daily quota exhausts quickly; auto-falls back to Emergent Universal Key
- ElevenLabs voice quota limited; falls back to Emergent-managed OpenAI Whisper/TTS
- Interactive MapLibre map does not render on browser/web preview (native-only); marker-list fallback shown on web
- No authentication system yet; fixed demo user_id used for all requests
- Push notifications (FCM) structured but not wired to a real Firebase project

12. Backend Test Coverage

- test_orca_backend.py — core chat/orchestrator pipeline
- test_pfz_recommendation.py — safest-PFZ recommendation logic
- test_pfz_regions.py — India-wide region resolution
- test_situation_and_route.py — /api/situation + route safety
- test_voice_situation_route.py — voice endpoints + situation route
- test_localization_manual.py — multilingual response verification

13. Deployment Notes

- Development: Metro bundler serves Expo web preview; FastAPI via Uvicorn on port 8001; MongoDB local
- Production build/publish via Emergent platform's Publish button (top-right) — generates iOS/Android builds
- Environment variables managed via .env files; never hardcoded
- Some native-only features (voice background playback, MapLibre WebView) require a real device/native build

14. Future Roadmap

- Live data source integration: replace mock adapters with real INCOIS, IMD, and Bhuvan feeds
- Push notifications via FCM for proactive hazard/cyclone alerts
- Phone OTP / account authentication for persistent user profiles
- Expand seeded region roster beyond the current 6 coastal reference points
- Restore native multilingual ElevenLabs voice once quota/key constraints are resolved

---

— End of Document —